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STRUCTURE AND METABOLISM OF LIPOPROTEINS

I/Introduction

- **Problem:** Lipids (whether from food or produced by the liver or adipose tissue) are **insoluble in water** → they **cannot be transported** directly into the blood (an aqueous environment).
- **Solution:** They bind to **specific proteins** to form lipoproteins → which are **soluble in water** → and can therefore **circulate in the plasma**.
- **Role of lipoproteins:** To act as 'carriers' of lipids between the intestine, liver, adipose tissue, etc., for: -utilisation (energy)
  -storage.

Important note:
Free fatty acids are an exception to this rule → they are **transported by albumin** (a plasma protein).

II/Structure of Lipoproteins

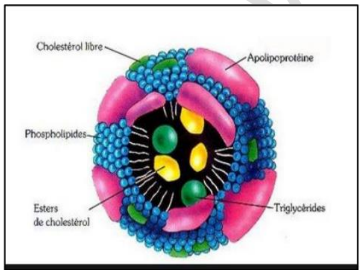
A. Definition: LP are **molecular aggregates** (heterogeneous assemblies) of specific transport **proteins**: Apo-lipoproteins with different combinations of **lipids** represented by: TG, phospholipids (PL), free cholesterol (CL) and esterified cholesterol (CE).

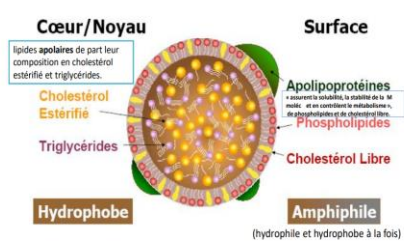
B. Composition:

- Hydrophobic core: TG + esterified cholesterol (CE).
- Outer membrane: Phospholipids (PL) + free cholesterol (CL) + apoproteins.

C. Cohesion:

- **Hydrophobic** bonds (between fatty acids and non-polar amino acids).
- **Ionic** bonds (polar groups of apoproteins and PL).





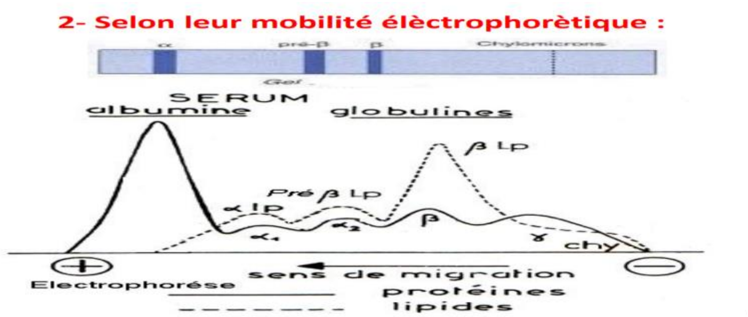
III. Classification of Lipoproteins

1. **By density:** Due to their lipid components, lipoproteins have a lower hydrated density than proteins, and this varies depending on the fraction (from least dense to most dense):

Lipoprotein	Density (g/mL)	Main physiological role
Chylomicrons	< 0.99	Transport of dietary TG
VLDL	0.99–1.006	Transport of endogenous TG
LDL	1.02–1.063	Transport of cholesterol to tissues
HDL	1.063–1.21	Transport of cholesterol to the liver

2. By electrophoretic mobility: In descending order of mobility, the following are found:

- Chylomicrons: **sediment (do not migrate)**
- VLDL: **pre-β.**
- LDL: **β.**
- HDL: **α.**



NB: Chylomicrons are not found in **the fasting** serum of a **healthy individual;** their **presence** indicates a **medica**l condition

IV. Origin of Circulating Lipids

- **Endogenous:** Hepatic synthesis (from glucose and free fatty acids).
- **Exogenous:** Dietary lipids hydrolysed by pancreatic lipase (with bile salts).

V. Apolipoproteins (Apo):

-Apo lipoproteins (Apo) are structural proteins essential for the formation and function of lipoproteins.

They play **four major** roles:

1. **Structure:** Maintaining the integrity of lipoproteins.
2. **Synthesis:** The formation and secretion of lipoproteins;
3. **Recognition**: They act as ligands for specific cell receptors.
4. **Regulation:** Activation or inhibition **of key enzymes** in lipid metabolism (e.g. lipoprotein lipase **(LPL)** or lecithin-cholesterol acyltransferase **(LCAT)).**

Main Classes and Subclasses

- **Apo A** (e.g. Apo A-I, A-II): Found mainly on HDL.
- **Apo B:** - Apo B-48: Found exclusively on chylomicrons.
  -Apo B-100: Found on VLDL, IDL and LDL.

- **Apo C** (C-I, C-II, C-III): **Transferable** between chylomicrons, VLDL and HDL.
  - -Apo C-II activates lipoprotein lipase (LPL).
  - -Apo C-III inhibeted lipoprotein lipase (LPL).
- **Apo D**: Associated with HDL.
- **Apo E** (E2, E3, E4): **Transferable**, involved in the hepatic uptake of remnants.
- **Others:** Apo L, M, O (less well-defined roles).

Key Functions

- Specificity: Each lipoprotein carries distinct apolipoproteins that determine its destination.

-Apo B-100 → Binds to the **LDL receptor**

-Apo E → Binds to **hepatic receptors.**

- **Transfer**: Certain Apo proteins **(such as Apo C and E)** can move from one lipoprotein to another to modulate their metabolism.

Example:

- **HDLs** rich in Apo A-I activate the enzyme LCAT for cholesterol esterification.
- **Chylomicrons** carry Apo B-48 for their intestinal synthesis and acquire Apo C-II (from HDLs) to activate LPL in peripheral tissues.

Apolipoprotein	**Carrying lipoprotein**	**Essential function**
Apo A-I / A-II	**HDL**	Activation of LCAT (anti-atherogenic)
Apo B-48	**Chylomicrons**	Intestinal synthesis
Apo B-100	**VLDL, IDL, LDL**	LDL receptor ligand
Apo C-II	**Chylomicrons, VLDL, HDL**	Activation of LPL
Apo E	**Chylomicrons, VLDL, HDL**	Liver recognition of remnants
Apo D	**HDL**	Transport of small molecules

Table: Percentage composition of lipoproteins in terms of lipids and apolipoproteins

	Triglycérides	Choléstérols libre	Cholestérols estérifiés	Phospholipides	Apolipoprotéines
Chylomicrons	85	1	2	8	B-48(C,E) 2
VLDL	55	7	10	20	B-100(C,E) 9
IDL	26	8	30	23	B-100 E 11
LDL	10	10	35	20	B100 20
HDL	8	5	15	25	AI(A,C,E,F,G,H) 45

VI. Lipoprotein Metabolism

A. Chylomicrons:

1. Formation

- **Origin:** Synthesised in enterocytes (intestinal cells).
- **Composition:**

-Triglycerides (TG): Derived from dietary lipids (the majority).

- Phospholipids and esterified cholesterol.

-Apolipoproteins: **Apo B-48 (specific),** Apo AI, Apo AIV.

2. Transport and Maturation

- Release: Enter the bloodstream via the lymphatic vessels.
- Maturation:

-Acquire Apo C and Apo E (derived from HDL).

-**Apo C-II** activates lipoprotein lipase (LPL).

3. Action in Peripheral Tissues

- **Role of LPL**: Hydrolyses TG into fatty acids (FA) and glycerol.

- Muscles: FA used for energy.

-Adipose tissue: FA stored as TG.

- **Result:**

-Loss of TG and Apo C.

-Conversion into chylomicron remnants (rich in cholesterol).

4. Fate of the Remnants

- Liver uptake: Via receptors recognising **Apo E.**
- Elimination: **Recycling** of components in the liver.

5. Main Role

- Transport of dietary (exogenous) triglycerides (TG) to tissues for:

6. Pathology:

A **defect in LPL** or **Apo C-II** leads to an accumulation of chylomicrons **(hypertriglyceridaemia)**

B. VLDL Metabolism

1. Synthesis and Composition

- **Origin**: Synthesised by **the liver.**
- **Composition:**

-Triglycerides **(TG):** Predominant (synthesised in the liver from fatty acids and glucose).
-Cholesterol: In free and esterified forms.
-Apolipoproteins: **Apo B-100** (essential) and **Apo E**.

2. Release and Maturation

- **Release:** Into the bloodstream.
- **Maturation**:

-Acquisition of **Apo C** (from HDL).
-Apo C-II activates lipoprotein lipase **(LPL).**

3. Action in Peripheral Tissues

- **Role of LPL**: Hydrolysis of TG into fatty acids (FA) and glycerol.
- **Result:**

-Loss of triglycerides **(TG)** and **apo C.**
-Conversion to **IDL** (intermediate-density lipoproteins).

4. Fate of IDLs

- **Main pathway:**
  -Uptake by the **liver** via LDL receptors (B/E) recognising **Apo B-100 and Apo E**.
  -Hepatic degradation.
- Secondary pathway:
  -Enrichment with esterified cholesterol **CE**(exchange with HDL via CETP).
  -Loss of **Apo E**.
  -Conversion to **LDL**.

5. Main Role

- Transport of endogenous **TG** (synthesised by the liver) to peripheral tissues.

6. Associated Conditions:

- Excess VLDL → **Hypertriglyceridaemia.**
- Excessive conversion to LDL → **Risk of atherosclerosis**.

C. LDL Metabolism

1. Formation of LDL

- **Origin:** Derived from **IDL** (which are themselves derived from VLDL) in the **bloodstream**.
- **Mechanism**:
  -Loss of Apo E and C.
  -Enrichment with **esterified cholesterol** via exchange with **HDL** (mediated by CETP).

2. Composition of LDL

- Lipids: Rich in **esterified cholesterol** (~50%).
- Apolipoprotein: **Apo B-100** (unique and essential for recognition).

3. Function and Fate

- **Transport**: Delivery of cholesterol to **peripheral tissues.**
- **Internalisation mechanism**:

a. Binding to **LDL receptors** (via Apo B-100).
b. Internalisation via endocytosis.
c. Degradation of the **LDL-receptor complex** in the lysosome:

- Recycling of **amino acids** (Apo B-100).
- Release of **cholesterol** for:
  - -Cell membranes.
  - -Synthesis of steroid hormones (cortisol, oestrogens, etc.).
  - -Storage (in the form of esters in lipid droplets).

4. Physiological Role

- **Cholesterol distribution**: From the liver to tissues that require it.
- **Regulation:**

Cells adjust the number of **LDL** receptors they express according to their cholesterol requirements (negative feedback).

5. Pathological Implications

- **Excess LDL**:

-Deposition in the arteries → Atherosclerosis.
-Main marker of cardiovascular risk.

- **Genetic defect** (e.g. **familial hypercholesterolaemia**): Non-functional LDL receptors → LDL accumulates in the blood.

Key points:
LDLs are the main **transporters of cholesterol to tissues**. An excess **of LDLs** is **atherogenic,** whilst their fine regulation is crucial for cellular homeostasis.

D. HDL Metabolism

1. Synthesis and Initial Structure

- **Origin:** Produced by the liver and intestine in the form of nascent (discoidal) HDL.
- **Initial Composition**:

-Apo AI and Apo AII (essential for function).
-Few lipids (mainly phospholipids).

2. Main Function: Reverse Cholesterol Transport

- **Step 1**: Cholesterol Uptake

HDLs interact with peripheral tissues via the ABCA1 receptor.
Export of free cholesterol from cells to HDLs.

- **Step 2**: Cholesterol Esterification

LCAT (activated by Apo AI) converts free cholesterol into esterified cholesterol (CE), stored in the core of HDL → Formation of HDL3 (spherical).

- **Step 3**: Exchange in Circulation

-HDL3 particles acquire Apo C/E and TG (via exchange with other lipoproteins, mediated by CETP).

-Conversion to HDL2 (larger, rich in CE and TG).

3. Final Destination in the Liver

- Option 1: Recycling into HDL3 after hydrolysis of TG by hepatic lipase.
- Option 2:

-Transfer of CE to hepatocytes via the SR-B1 (scavenger) receptor.

-Cholesterol is excreted in bile (as bile acids or in free form).

4. Key Roles of HDL

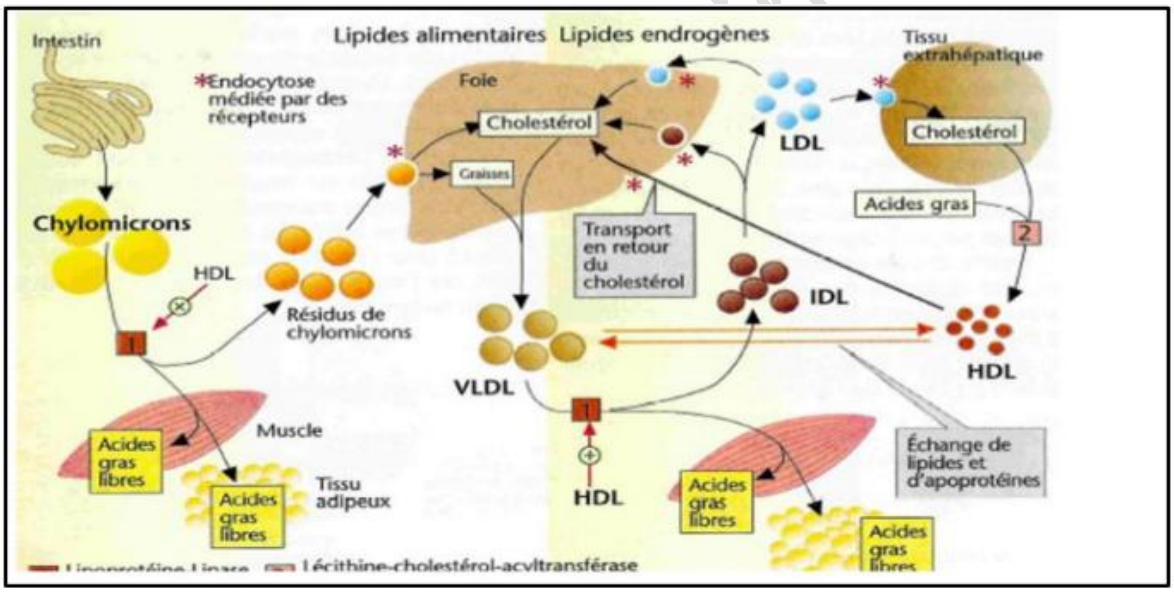
1. Reverse cholesterol transport: Clears excess cholesterol from tissues **(anti-atherogenic).**
2. Transports 25% of plasma cholesterol.
3. Reservoir of apolipoproteins (Apo C, E) for other lipoproteins.

5. Medical Implications

- **High HDL**: Associated with a reduced cardiovascular risk ("good cholesterol").
- **Low HDL:** Linked to atherosclerosis.

Key Points:

HDL acts as **a vascular cleaner**, recycling cholesterol back to the liver and thereby protecting the arteries. Their function depends closely on **Apo AI** and **LCAT**.

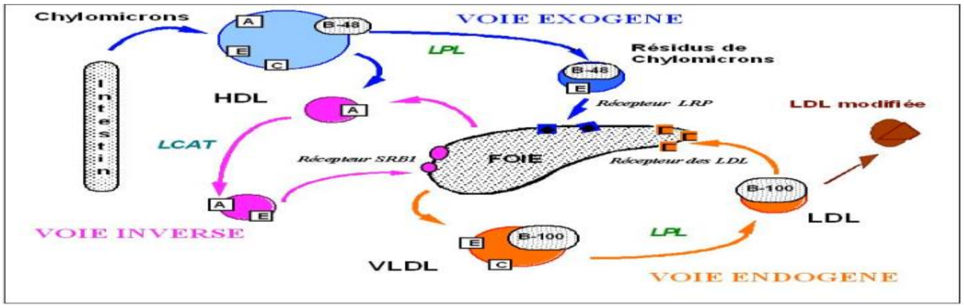


A summary diagram of the metabolism of dietary lipids.

System open to the environment

Three main regulatory pathways

Exogenous/ Endogenous/ Inverse



VII. Summary Diagram

There are three interconnected metabolic pathways:

1. **Exogenous pathway** (dietary lipids):

Intestine → Chylomicrons → LPL → Remnants → Liver

2. **Endogenous pathway** (hepatic cholesterol):

Liver → VLDL → LPL → IDL → LDL → Peripheral tissues

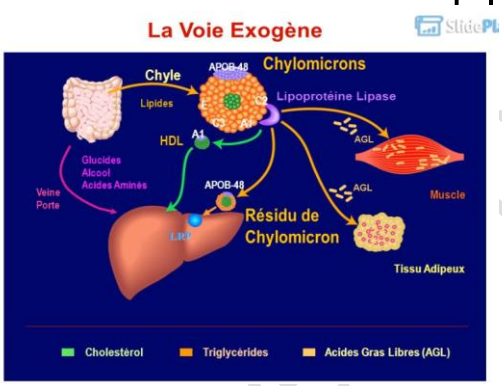
3. **Reverse cholesterol transport:**

Tissues → Nascent HDL → LCAT → Mature HDL → Liver → Bile

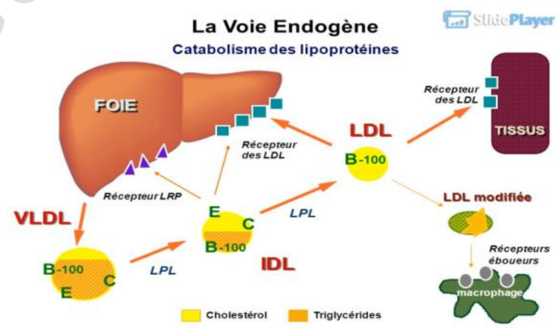
Associated conditions:

- Excess LDL → Atherosclerosis.
- Low HDL → Increased cardiovascular risk.

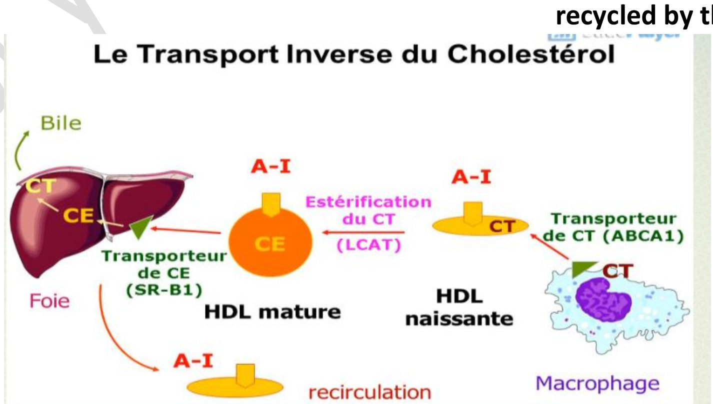
Lipoprotein metabolism



The distribution of lipids of intestinal origin



The distribution of lipids synthesised or recycled by the liver



The return of cholesterol to the liver